HDOC- 10 – 1
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Program to Display the Day of the Week from a Number (1–7) Using switch-case

This document provides a clean, well-commented implementation for displaying the day of the week from a number in C. The program takes a number between 1 and 7 as input and uses a switch-case statement to print the matching day, with a default case for invalid numbers.

1. C Implementation

Step 1: Understand the Logic

- The numbers 1 to 7 are mapped to the days as: 1 = Monday, 2 = Tuesday, 3 = Wednesday, 4 = Thursday, 5 = Friday, 6 = Saturday, 7 = Sunday.
- A switch statement compares the value of one variable with the value of each case label and runs the matching case.
- The break statement at the end of each case stops the program from running into the next case (fall-through).
- The default case runs when the number does not match any case, so it is used to report invalid input.

Step 2: Define Variables and Data Types

- Use an integer variable (int) to store the day number.

Step 3: Take User Input

- Display a prompt using printf("Enter a day number (1-7): "), then read the value using scanf("%d", &day).

Step 4: Apply the switch-case Statement

1. Write switch (day) so that the value of day is compared with each case.
2. For case 1 to case 7, print the matching day name and end the case with break.
3. In the default case, print an error message for any other number.

Step 5: Format the Output

- Print the name of the day using printf(), or the error message if the number is not between 1 and 7.

Complete C Code

```c
/*
 * Q15 (User Inputs, Operations & Output)
 * Write a program to display the day of the week based on
 * a number (1-7) using switch-case.
 */
#include <stdio.h>

int main() {
    int day;

    // Step 1: Read the day number
    printf("Enter a day number (1-7): ");
    scanf("%d", &day);

    // Step 2: Display the day using switch-case
    switch (day) {
        case 1:
            printf("Monday\n");
            break;
        case 2:
            printf("Tuesday\n");
            break;
        case 3:
            printf("Wednesday\n");
            break;
        case 4:
            printf("Thursday\n");
            break;
        case 5:
            printf("Friday\n");
            break;
        case 6:
            printf("Saturday\n");
            break;
        case 7:
            printf("Sunday\n");
            break;
        default:
            printf("Invalid input! Enter a number between 1 and 7.\n");
    }

    return 0;
}
```

Program Output (Sample Runs)

Run 1:

```
$ gcc hdoc10_1.c -o hdoc10_1
$ ./hdoc10_1
Enter a day number (1-7): 1
Monday
```

Run 2:

```
$ ./hdoc10_1
Enter a day number (1-7): 5
Friday
```

Run 3:

```
$ ./hdoc10_1
Enter a day number (1-7): 7
Sunday
```

Run 4:

```
$ ./hdoc10_1
Enter a day number (1-7): 9
Invalid input! Enter a number between 1 and 7.
```

Dry Run with Test Cases

Sample Test Case 1:

- Input: 1
- Calculation:
  - o switch (day) compares day = 1 with the case labels
  - o It matches case 1, so printf("Monday") runs
  - o break exits the switch, so no other case runs
- Output: Monday

Sample Test Case 2:

- Input: 5
- Calculation:
  - o switch (day) compares day = 5 with the case labels
  - o Cases 1, 2, 3 and 4 do not match; it matches case 5, so printf("Friday") runs
  - o break exits the switch, so no other case runs
- Output: Friday

Sample Test Case 3:

- Input: 7
- Calculation:
  - o switch (day) compares day = 7 with the case labels
  - o Cases 1 to 6 do not match; it matches case 7, so printf("Sunday") runs
  - o break exits the switch
- Output: Sunday

Sample Test Case 4:

- Input: 9
- Calculation:
  - o switch (day) compares day = 9 with the case labels
  - o None of the cases 1 to 7 match 9
  - o The default case runs and prints the error message
- Output: Invalid input! Enter a number between 1 and 7.